RESCUELOGIC®

SAFER Protocol FAQ

Security And Fire Emergency Reports

Version 1.1 • October 2026

CORE MESSAGE SAFER standardizes life-safety event meaning, not life-safety systems. Transport moves the message. SAFER defines what the message means.

PAYLOAD ONLY SAFER defines the structure of a reported event payload. It does not define pathway supervision, network architecture, transport security, product listing, NFPA 72 compliance, UL evaluation, supervising-station requirements, or emergency response policy.

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Companion to the proposed SAFER Protocol v1.1

How to use this FAQ

This FAQ is written for manufacturers, integrators, monitoring platforms, software developers, fire-alarm professionals, authorities, and reviewers who need to understand what SAFER does, what it does not do, and how it can be tested without disrupting existing life-safety obligations.

REVISION EMPHASIS Version 1.1 clarifies SAFER's purpose as a payload structure for reporting event meaning. It preserves the seven required fields and six STATUS codes of v1.0 while explaining how SAFER can coexist with, wrap, translate, or reference legacy formats such as Contact ID. Proposed changes to the representation of condition and event qualifier remain open for future discussion.

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24–26	Testing and companion documents: what to test first and which supporting documents belong in which context.

SAFER at a glance

Item	Plain-language answer
What SAFER is	A structured, JSON-based payload format for reporting life-safety and security events.
What SAFER standardizes	The meaning of a reported event: source context, point identity, device type, location, and defined reporting status, with optional descriptive condition detail.
What SAFER does not standardize	The life-safety system, pathway supervision, transport method, listing, certification, regulatory compliance, or response policy.
Adoption posture	Coexist • Wrap • Translate • Reference • Migrate
Core sentence	This system reports that this device, of this type, at this location, has changed to this state.

Frequently asked questions

1. What is SAFER?

The Security And Fire Emergency Reports (SAFER) Protocol defines a structured, JSON-based event payload for reporting life-safety and security events.

SAFER gives alarm systems, monitoring platforms, software applications, and people a common way to understand the basic meaning of a reported event without requiring every system to use the same internal architecture.

CORE SENTENCE This system reports that this device, of this type, at this location, has changed to this state.

2. Why was SAFER created?

Many alarm reporting formats were designed when bandwidth and processing power were scarce. Compact codes solved an important problem, but they often require receiving systems to reconstruct meaning from code tables, local databases, or proprietary context.

SAFER was created so event information can travel as structured meaning: who or what reported, which point was involved, what kind of point it was, where it belongs, and what condition was reported.

- Reduce information loss when events cross system boundaries.
- Preserve human-readable device and location context.
- Make event messages easier for software, analytics, dashboards, and AI-assisted tools to interpret.
- Allow gradual migration while preserving compatibility with existing systems.

3. What does “SAFER standardizes life-safety event meaning, not life-safety systems” mean?

It means SAFER is not a replacement for the fire alarm system, supervising station, communication pathway, listed product, or response procedure.

SAFER defines a payload structure that carries event meaning. Other codes, standards, manufacturers, and project requirements remain responsible for system design, reliability, listing, supervision, and response.

ADOPTION PRINCIPLE SAFER is a common event language. It is not a substitute for listed equipment, code compliance, or qualified design.

4. What does “Transport ≠ Meaning” mean?

Modern transport technologies can deliver messages quickly and reliably, but delivery alone does not tell the receiving system what the message means.

A message can be carried by MQTT, HTTP, AMQP, WebSockets, a database, a file, or another pathway. SAFER defines the event payload meaning that can travel inside or across those technologies.

Question / item	Answer / meaning
Transport asks	Can the message be delivered, routed, acknowledged, logged, or queued?
Meaning asks	Can the receiving system understand the reported event consistently?
SAFER’s role	Define the event meaning carried by the payload.

5. Why does SAFER standardize only the event payload?

Life-safety infrastructure involves critical responsibilities that are already governed by established codes, standards, listings, manufacturer instructions, and project requirements.

SAFER therefore focuses on a narrower problem: how to represent a reported event as information.

- Pathway supervision remains outside SAFER.
- Transport security remains outside SAFER.
- Product listing and certification remain outside SAFER.
- Emergency response policy remains outside SAFER.
- The event payload structure is the part SAFER defines.

6. What does SAFER define, and what does it not define?

The safest way to describe the scope is with a boundary table.

Question / item	Answer / meaning
SAFER defines	Event payload fields; event-status vocabulary; versioned message identity; optional structures that can carry additional context.
SAFER does not define	Pathway supervision; network architecture; product listing; regulatory approval; NFPA 72 compliance; UL evaluation; supervising-station rules; emergency response policy.
Implementers remain responsible for	Applicable codes, standards, equipment listings, cybersecurity, transport choices, reliability, testing, and authority requirements.

7. What is a SAFER event?

A SAFER event is a structured report of a condition involving a known point or device. In plain language:

> **PRIMITIVE EVENT STATEMENT** This system reports that this device, of this type, at this location, has changed to this state.

Question / item	Answer / meaning
System / source context	Who is reporting?
Identity	Which device or point?
Device Type	What kind of thing is it?
Location	Where does it belong?
Reported condition	What condition is the source describing?

The five-role model describes the information associated with a change of condition. A transmitted SAFER report may also describe restoration, a repeated condition, or a status-only observation. SAFER STATUS is the numeric reporting code, not necessarily the current persistent state of a device.

8. Why does SAFER separate device information from device state?

A point has an identity within its reporting source context. DEVICE_ID identifies it; DEVICE_TYPE and DEVICE_LOCATION describe what it is and where it belongs. These descriptions may be corrected without making it a different point. The conceptual event model describes a change of condition; the transmitted SAFER STATUS field is a reporting code.

A report may describe activation, trouble, supervisory, restoration, a repeated condition, or status-only information. Restore and previously reported describe reporting relationships, not independent persistent states of the device.

USEFUL DISTINCTION DEVICE_ID identifies the point within its reporting context. DEVICE_TYPE and DEVICE_LOCATION describe that point. SAFER STATUS communicates the defined reporting meaning, while the receiving application may use source context and event history to determine the displayed state.

9. Why does device type matter?

The same underlying change can mean different things when it belongs to different objects. A dry contact operating on a manual pull station and a dry contact operating on a sprinkler control-valve switch may look similar electrically, but they do not mean the same thing operationally.

OBJECT + CONDITION Condition alone does not carry all meaning. Device type and site configuration help determine classification and response.

Question / item	Answer / meaning
Manual pull station + active condition	Fire-alarm interpretation
Valve supervisory switch + active condition	Supervisory interpretation
Monitoring path fault	Trouble or integrity problem

10. Why is STATUS represented as a number?

Numeric STATUS values allow receiving software to interpret reports without depending on manufacturer-specific or localized display wording.

SAFER v1.1 retains six numeric reporting codes. Some describe conditions or classifications, while others describe restoration, report purpose, or reporting history. These meanings are not necessarily independent physical states.

The optional CONDITION field can preserve a more specific human-readable description. For example, several reports can share SAFER STATUS=4 (Trouble) while describing Open Circuit, Ground Fault, or Detector Missing.

RescueLogic demonstrates one way to present this information. Its three Base States—Normal, Alarm, and Trouble—help people recognize a situation through familiar device images, while descriptive Condition text gives them additional information about what the source reported.

A small number of recognizable state categories can support a much larger variety of descriptive conditions.

SAFER v1.1 keeps its six reporting codes. Question 26 explores whether a future version should distinguish a reported condition from how it was reported.

STABLE MACHINE VALUE An HMI may display ACTIVE, Alarm, Alarme, or another localized label while the transmitted STATUS value remains stable.

Question / item	Answer / meaning
1	Active
2	Supervisory
3	Restore
4	Trouble
5	Status only
6	Previously reported

11. Why does SAFER use JSON?

JSON is widely supported, human-readable, machine-readable, and easy to validate. It is already common in web services, cloud platforms, IoT systems, databases, and developer tools.

Using JSON allows SAFER to carry more context than compact legacy reporting formats while remaining familiar to modern software environments.

- Readable by people during troubleshooting.
- Easy for machines to parse and validate.
- Supported by nearly every programming language.
- Extensible through optional fields and versioned schemas.

12. Why are SAFER field names written in ALL CAPS?

Uppercase field names make the SAFER vocabulary visually distinct when it appears beside ordinary application data.

This mirrors the visual tradition of alarm-panel displays, point identifiers, technical labels, and drawings, while giving software a clear set of defined field names to test.

> **NAMING LAYER** Uppercase is a representation convention. Meaning comes from the defined SAFER field names and schema.

13. How is SAFER different from Contact ID?

Contact ID is a compact reporting format designed for constrained signaling environments. It has served the alarm industry well, and SAFER does not need to attack it to explain why richer structured payloads are useful.

SAFER carries reporting source, point identity, device type, location, STATUS, message identity, and optional source records. Contact ID event codes and Q qualifiers should be interpreted together with account and partition-zone/user context, not treated as exhaustive lists of present and future device types.

CONTACT ID QUALIFIER — Q=1: new/opening. Q=3: restore/closing. Q=6: previously reported condition still present. An event code, qualifier, and partition-zone/user identification have separate jobs; Q=6 does not mean TROUBLE. Contact ID is a message format, not the physical transport.

```
Contact ID example:
1234 18 1101 01 015 8

SAFER example:
{
  "CUSTOMER": "1234",
  "PANEL_ID": "Node 01",
  "DEVICE_ID": "01-015",
  "DEVICE_TYPE": "Smoke Detector",
  "DEVICE_LOCATION": "Front Lobby",
  "STATUS": 1,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "123e4567-e89b-12d3-a456-426614174000"
  }
}
```

The original Contact ID message is not designed to contain the device type or room description. In this illustrative translation, those details come from known point records or verified configuration, not from the Contact ID message itself.

14. Is SAFER intended to replace existing alarm communication standards?

No. SAFER is designed to coexist with existing alarm communication methods and can support gradual migration.

The preferred adoption language is:

COEXIST • WRAP • TRANSLATE • REFERENCE • MIGRATE SAFER can coexist with legacy formats, wrap or translate legacy reports, preserve legacy references, and support gradual migration toward richer structured event reporting.

15. Can legacy alarm formats be translated into SAFER?

Yes, when source information and reliable site configuration establish the required SAFER fields. Translation is not a universal one-to-one mapping from a legacy event code to STATUS.

For Contact ID, Q=1 means new event/opening; Q=3 restore/closing; Q=6 previously reported condition still present. These qualifier numbers do not equal SAFER STATUS numbers. Preserve the original code, qualifier, and partition-zone/user context with the translated report where available.

- Establish both CUSTOMER and PANEL_ID from the source report or documented source/site configuration. Do not invent missing values.
- Translate the point or contact value into DEVICE_ID where mapping is available.
- Obtain device type and location from known source records or verified configuration; identify looked-up or derived facts and do not invent missing information.
- Preserve the original Contact ID string as an optional reference when useful.

16. Can SAFER work with existing fire alarm systems?

Yes. Most fire alarm systems already contain the essential information internally: point identity, device type, location text, and event state.

SAFER provides a structured way to represent that information when it is exported, integrated, logged, displayed, translated, or transmitted to another system. Existing systems can continue operating normally while SAFER is tested at an integration or reporting boundary.

LOW-DISRUPTION PATH A system can begin by producing or consuming SAFER payloads at an interface without redesigning its listed core functions.

17. Would SAFER affect UL, NFPA, or code compliance?

SAFER by itself does not alter product listing, supervision requirements, pathway classifications, installation requirements, or code compliance.

If SAFER is implemented inside a product or code-required system function, that implementation may need to be evaluated under the applicable product standards, project requirements, and authority review. The SAFER specification itself remains an event-payload structure.

COMPLIANCE BOUNDARY SAFER describes event representation. It does not constitute a listed product, approved system configuration, or compliance determination.

18. Is SAFER reliable enough for life-safety systems?

Reliability in life-safety systems comes from the listed equipment, approved system design, supervised pathways, installation, testing, maintenance, and applicable codes and standards.

SAFER does not make a system reliable or unreliable by itself. SAFER addresses the clarity and interoperability of event information once that information is represented as a payload.

- The payload can be carried by many infrastructures.
- The infrastructure must still meet applicable reliability and supervision requirements.
- SAFER helps preserve meaning; it does not replace reliability engineering.

19. Does SAFER follow established network architecture principles?

Yes. SAFER separates the information carried by the event from the transport mechanism used to move it.

That separation aligns with a common network-design idea: communication systems often work best when transport and application meaning are kept in distinct layers. The network moves information; endpoints and applications interpret it. SAFER defines the structured event meaning that the receiving endpoint can interpret.

TRANSPORT VS MEANING A transport can successfully deliver a payload without defining what that payload means. SAFER supplies the life-safety event meaning.

20. How does SAFER relate to modern cloud and IoT monitoring systems?

Modern cloud and IoT systems often describe a known entity and its condition. SAFER uses a similar information pattern, while distinguishing a physical change of condition from a report that may be new, restored, repeated, or status-only.

DEVICE_ID, DEVICE_TYPE, and DEVICE_LOCATION describe a point in its reporting context. SAFER STATUS remains the versioned reporting code; it should not be equated to a complete, persistent STATE model.

SIMPLE MODEL DEVICE = (ID, TYPE, LOCATION)
REPORT CODE = SAFER STATUS
EVENT = ΔSTATE × DEVICE × TIME
The expression for EVENT describes a conceptual change of state. A transmitted report may also describe restoration, repetition, or an observation without establishing that a new physical transition occurred.

21. Why not just use MQTT, AMQP, HTTP, or WebSockets?

Those technologies move messages. They do not, by themselves, define the life-safety meaning inside the message.

SAFER can be carried by MQTT, AMQP, HTTP, WebSockets, files, APIs, databases, or other mechanisms. SAFER is the payload grammar; the surrounding technology is the carrier.

Question / item	Answer / meaning
MQTT / AMQP / HTTP / WebSockets	Transport, routing, request/response, queuing, or delivery behavior.
SAFER	Payload meaning for a reported life-safety or security event.
Together	A transport carries a SAFER event payload that receiving systems can interpret consistently.

22. Is SAFER similar to SNMP?

There is a useful conceptual similarity: both approaches allow systems from different manufacturers to report operational information in a structured way.

The difference is scope. SNMP is a broader network-management protocol. SAFER focuses on the structure of reported life-safety and security event information. SAFER does not define how devices are queried, how pathways are supervised, or how management operations are performed.

ANALOGY, NOT IDENTITY SNMP helps normalize network-management information. SAFER helps normalize life-safety event meaning.

23. Is SAFER intended to become a standard?

SAFER is currently published as an open specification and reference structure. It is not presented as a mandated code requirement.

If the structure proves useful in implementation, standards organizations, industry groups, manufacturers, integrators, or public-safety stakeholders may choose to evaluate similar concepts for formal standardization or adoption.

PRACTICAL PATH Build, test, learn, and refine the event model. Formal recognition can follow demonstrated usefulness.

24. Why publish SAFER under the Apache License 2.0?

Life-safety interoperability benefits from transparency and collaboration. Publishing SAFER under a permissive open license lowers barriers for manufacturers, monitoring platforms, integrators, researchers, and developers who want to test the structure.

The goal is not to lock up the vocabulary. The goal is to encourage experiments, mappings, validators, sample payloads, translators, and interoperable implementations.

- Developers can test without negotiating a private license first.
- Manufacturers can evaluate the payload structure as an implementation option.
- Researchers can review and critique the model.
- Industry participants can suggest improvements and extensions.

25. What should developers or integrators test first?

Start with a known source report and check which fields are reported, referenced, derived, or unknown. A first test can be a translator, validator, dashboard, or event log; it need not redesign a life-safety system.

A useful extra test is a repeated supervisory report: confirm that the supervisory classification is not lost when the same condition is reported again. Also test an uncorrelated restore and missing device location; absence of information is not proof of NORMAL.

TESTING OBJECTIVE Demonstrate that different systems can understand the same event meaning without forcing them to share the same internal architecture.

- Validate a minimal SAFER payload against the v1.1 schema. Reject a version other than "1.1" or an empty SAFER_ID.unique_id.
- Map one known panel event into CUSTOMER, PANEL_ID, DEVICE_ID, DEVICE_TYPE, DEVICE_LOCATION, STATUS, and SAFER_ID.
- Preserve a legacy reference such as Contact ID inside an optional field.
- Render the same event in a dashboard, log, and floor-plan view without changing the source facts.
- Test conflicting records and preserve uncertainty rather than forcing a false answer.
- Compare two reports with the same SAFER STATUS but different CONDITION descriptions, such as Open Circuit and Ground Fault. Confirm that the application can preserve the distinction even if it displays both reports using the same Trouble state or device image.

26. What design question remains open?

SAFER v1.1 preserves six STATUS codes for compatibility and simplicity. However, these codes combine meanings from different dimensions of reporting. For example, a supervisory condition can also be previously reported.

Two legitimate concerns deserve consideration: preserving a compact, familiar reporting vocabulary, and maintaining sufficient precision to represent condition, classification, and report relationship independently.

Examples showing how reported conditions relate to new, restored, repeated, or status-only messages are welcome through GitHub Issues. Please include the original reporting format, expected interpretation, information that might be lost, and compatibility concerns. Suggested changes can be discussed and tested before consideration for a future SAFER version. Version 1.1 retains its six STATUS codes; incompatible changes follow the major-version rules.

Companion documents and recommended contexts

Choose a companion based on what you want to understand next.

Document	Best role	Recommended context
README.md	GitHub landing page	First impression for developers, reviewers, and public visitors. Short, scannable, and test-oriented.
SAFER Protocol v1.1	Authoritative protocol specification	Primary specification for fields, JSON structure, versioning, optional elements, and compliance boundaries.
SAFER FAQ v1.1	Adoption and objection handling	For manufacturers, integrators, AHJs, legal/compliance readers, and anyone asking what SAFER does or does not change.
SAFER Common Vocabulary	Shared terminology	Defines SAFER STATUS, optional CONDITION, Contact ID qualifiers, RescueLogic Base States, and future design proposals; companion, not a change to v1.1.
A Minimal Event Information Model — Review Version	Technical research/review paper	For IoT architects, standards reviewers, academic/research reviewers, NFPA 950-style data-exchange discussions, and semantic-interoperability review.
What Is an Alarm Event?	One-page field/training handout	Good for technicians, sales engineers, construction trailers, orientation, posters, and plain-language explanations.
Fire Alarm Panels Speak the Same Language	Visual proof / evidence sheet	Shows the common grammar across manufacturers. For presentations, appendices, and early conversations.
The Grammar of a Fire Alarm System	Plain-language educational companion	Best for non-developers who need the conceptual model before the protocol details.
What Does Everybody Know?	Plain-language discovery essay	Shows how understanding different manufacturer reports made the common information structure visible.
The Ordered Fire Alarm System	RescueLogic operations and product companion	Best for customers, technicians, and product users. Product-dependent uses of SAFER available separately.
End-of-Line vs. End-to-End	NFPA 72 / pathway terminology education	Use only in pathway, Class C/N, end-to-end, or network-supervision conversations. (Not in the first SAFER adoption packet.)
One Architecture. Many Manufacturers. Decades of Continuity.	Engineering history	Dan Horon's firsthand history of manufacturer interfaces, a common installation model, verification, and preservation.

SUGGESTED FIRST PUBLIC PACKET README.md, What Does Everybody Know?, SAFER Protocol v1.1, SAFER FAQ v1.1, SAFER Common Vocabulary, JSON schema, a few example payloads, What Is an Alarm Event?, and Fire Alarm Panels Speak the Same Language. Keep RescueLogic-specific product materials separate until the reader asks how the model becomes workflow or software.

Version 1.1 FAQ revision notes

- Emphasizes “SAFER standardizes life-safety event meaning, not life-safety systems.”
- Clarifies Transport ≠ Meaning and Payload Only as the two main adoption concepts.
- Reframes Contact ID language around coexistence, wrapping, translation, reference, and migration.
- Adds stronger NFPA/UL/listing/supervision boundary language.
- Adds a first-test path for developers and integrators.
- Adds guidance on companion documents and when to use them.
- Explains the distinction among numeric SAFER STATUS, optional CONDITION, and human-facing RescueLogic Base States.
- Keeps the condition/qualifier discussion open without altering v1.1 conformance.

Questions, feedback, and stewardship

Questions, implementation feedback, sample payloads, schema suggestions, and mapping experience may be directed to:

Dan Horon — dan@rescuelogic.com

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GitHub discussion and source documents:
https://github.com/rescuelogic/SAFERhttps://github.com/rescuelogic/SAFER

PUBLICATION NOTE Publication of this FAQ or specification does not imply endorsement by NFPA, UL, ULC, or any standards development organization. Project requirements should be verified against the applicable adopted codes, listed equipment, manufacturer documentation, and AHJ requirements.